

Thermodynamic Equilibrium & Trophic Cascade Architecture in Web of Life: Sprint 026

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Web of Life Simulation Engine

Sprint 026 Preprint – Repository:
<https://github.com/pascalranoroarijaona/WebOfLife>

Abstract

Sprint 026 establishes the rigorous enforcement of thermodynamic laws across the Web of Life simulation engine. Moving beyond structural population dynamics, this sprint details class hierarchies, monad stock state transformations, and interface contracts required to model multi-trophic cascades, metabolic heat dissipation, and strict solar-only energy influx. We formalize energy conservation ($\Delta E_{sys} = 0$) and atomic mass balance (Carbon, Nitrogen pools) alongside Second Law entropy generation ($\Delta S_{gen} \geq 0$). Executable Python monad pipelines demonstrate deterministic state transitions across primary producers, consumers, and environmental detrital sinks.

1 Introduction and Thermodynamic Framework

The **Web of Life** simulation models complex adaptive ecosystems as closed thermodynamic systems bounded by a single external exergy source: electromagnetic radiation flux from a ‘SolarSource’ singleton.

1.1 First and Second Laws of Thermodynamics

1. **Energy Conservation (1st Law):** Total system energy (E_{sys}) is conserved across all transitions:

$$\Delta E_{sys} = E_{solar_in} - E_{dissipated} = 0$$

Matter (carbon, nitrogen pools) must be strictly conserved; atoms cannot be created or destroyed.

2. **Entropy Generation (2nd Law):** Every metabolic transaction, movement, and predation event incurs an entropy tax, releasing thermal energy (Q) into the environmental sink:

$$\Delta S_{gen} = \frac{\dot{Q}_{dissipated}}{T_{env}} \geq 0$$

2 Class Hierarchy and Interface Contracts

We extend the object-oriented architecture to support multi-trophic interactions and rigorous material/energy tracking:

- **Abiotic:** SolarSource, Atmosphere, SoilMatrix
- **Biotic:** Organism (Abstract Base) $\rightarrow$ Autotroph (Plants/Algae) & Heterotroph (Herbivores, Carnivores, Detritivores).

Core interface contracts enforce strict typing:

- `IThermodynamicSystem`: `get_energy_stock() → float`, `dissipate_heat(joules: float) → None`
- `IMaterialPool`: `transfer_mass(target, stoichiometry) → bool`

3 Process Parameters and Governing Equations

All biochemical transitions adhere to strict stoichiometry and conservation laws defined in Table 1.

Table 1: Physical & Biochemical Process Parameters

Parameter	Symbol	Value / Range	Units	Description
Photosynthetic Efficiency	η_{photo}	0.015 – 0.035	dimensionless	Solar PAR to chemical bond c
Herbivore Assimilation	ϵ_{herb}	0.30 – 0.50	dimensionless	Plant biomass assimilation frac
Carnivore Assimilation	ϵ_{carn}	0.70 – 0.90	dimensionless	Prey biomass assimilation frac
Carbon Fraction	C_{frac}	0.47	kg C/kg Biomass	Dry-weight carbon content
Nitrogen Fraction	N_{frac}	0.07	kg N/kg Biomass	Dry-weight nitrogen content
Basal Metabolic Coeff.	b_m	0.001	J/(kg · s)	Resting heat dissipation scalar

3.1 State Transition Equations

1. Autotroph Production:

$$E_{stored}(t + dt) = E_{stored}(t) + (\text{Solar Flux} \times \eta_{photo} - \text{Respiration}) \cdot dt$$

2. Heterotroph Trophic Transfer:

$$\Delta E_{ingested} = \epsilon_{assimilation} \cdot E_{prey_consumed}$$

$$\text{Heat Dissipated} = (1 - \epsilon_{assimilation}) \cdot E_{prey_consumed} + \text{Basal Cost}$$

4 Verification and Invariants

Automated unit tests assert that total carbon and nitrogen across `SoilMatrix` and all organisms remain invariant within $\epsilon = 10^{-9}$ across 10,000 simulation steps, while guaranteeing $\Delta S \geq 0$ for every local tick.

5 Conclusion

Sprint 026 successfully embeds thermodynamic rigor and trophic cascade mechanics into the Web of Life engine, establishing a mathematically sound foundation for future evolutionary and behavioral simulations.

Official Repository: <https://github.com/pascalranoroarijaona/WebOfLife>